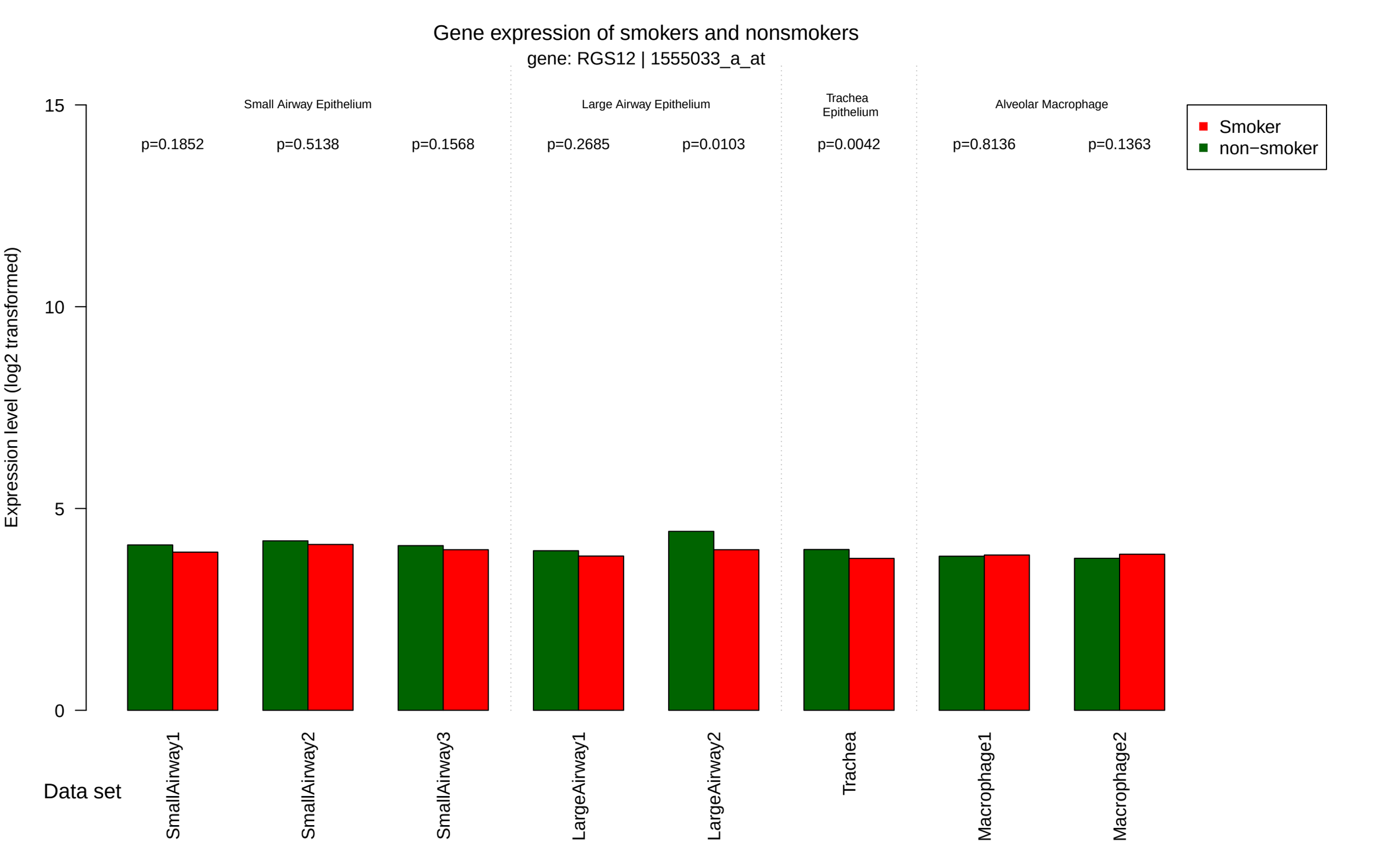
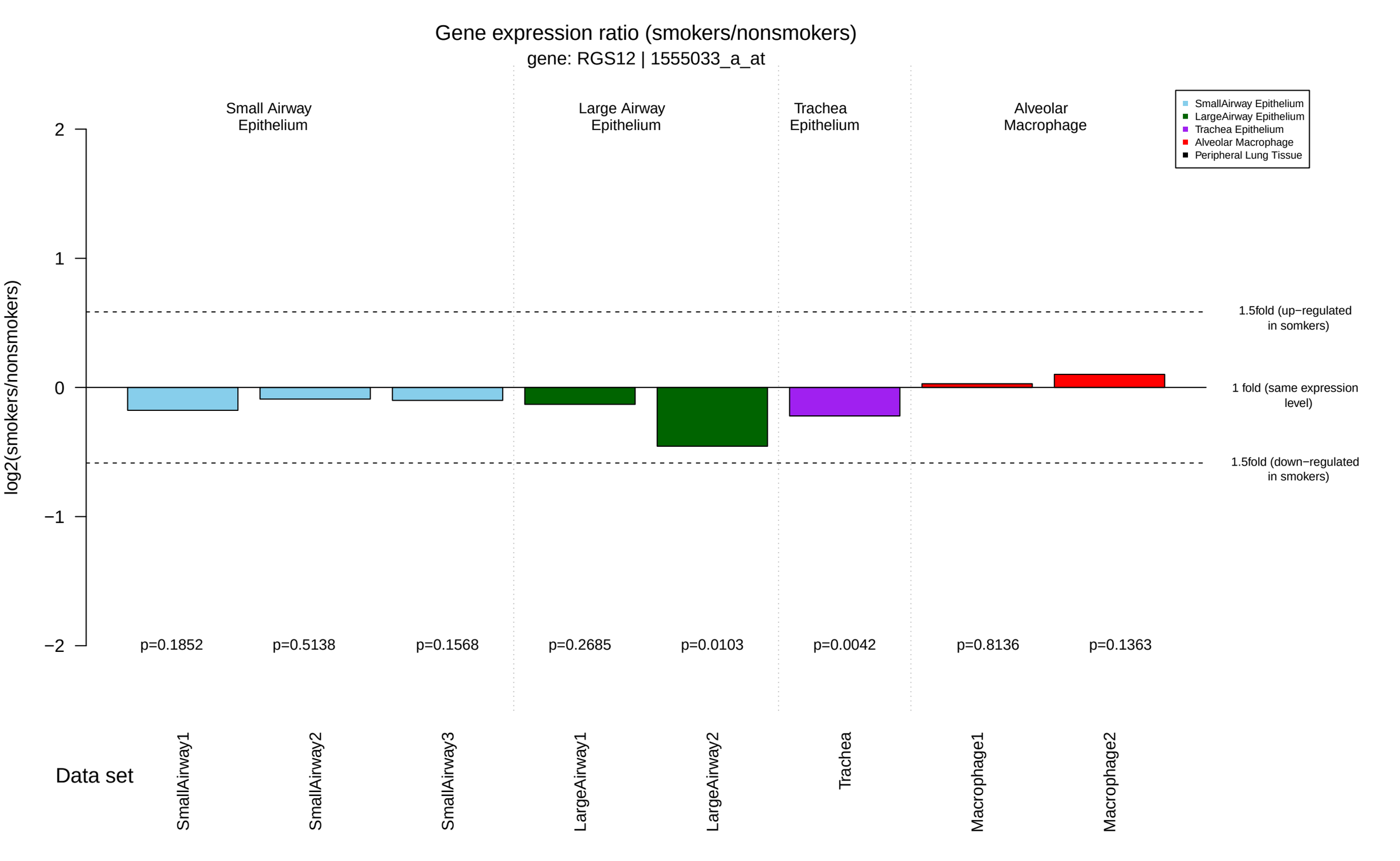
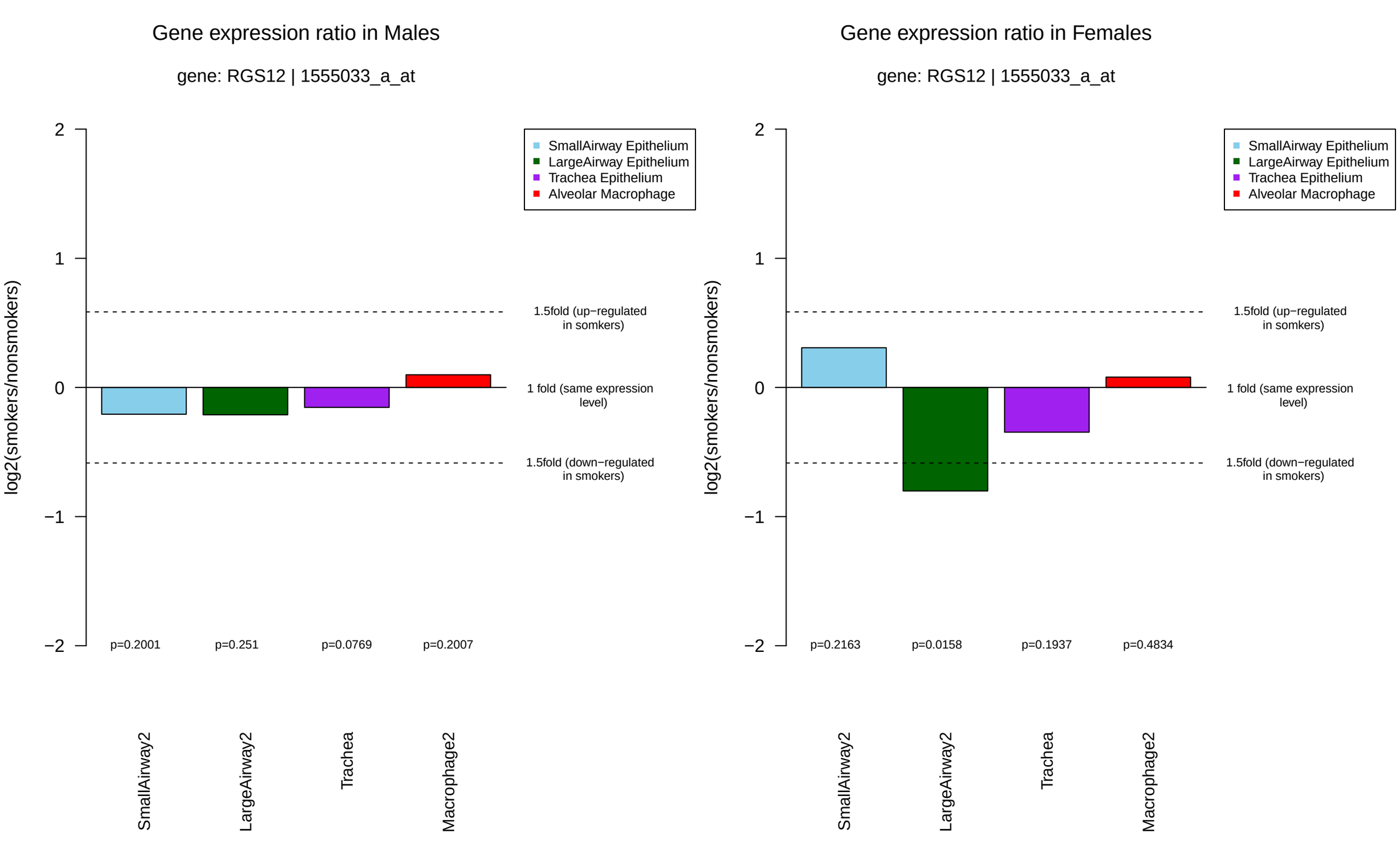
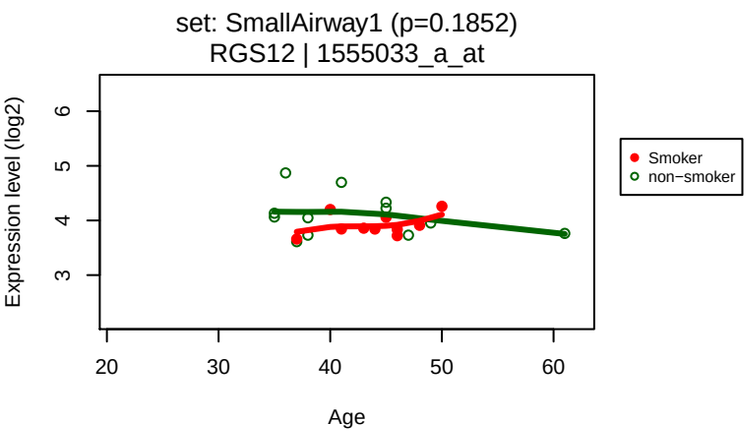
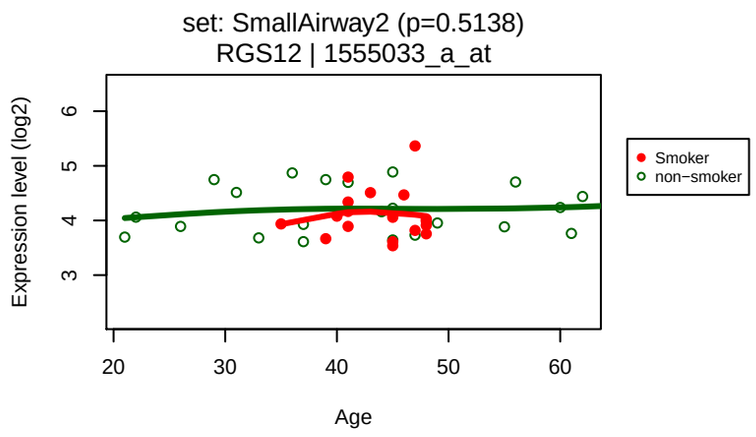
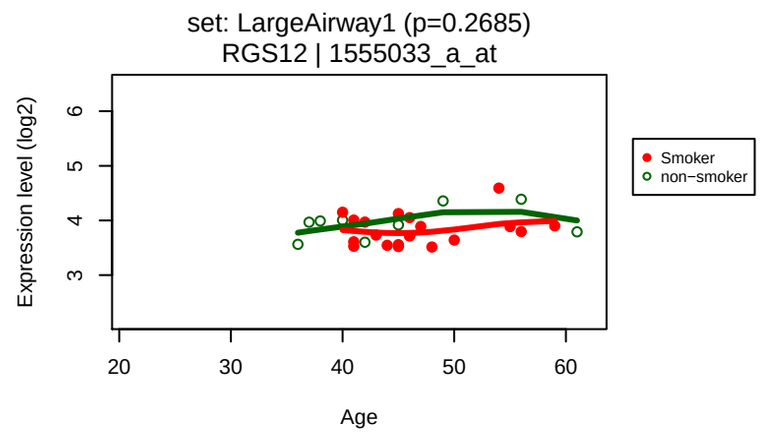

Gene expression of smokers and nonsmokers
gene: RGS12 | 1555033_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1852
p=0.5138
p=0.1568
p=0.2685
p=0.0103
p=0.0042
p=0.8136
p=0.1363
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

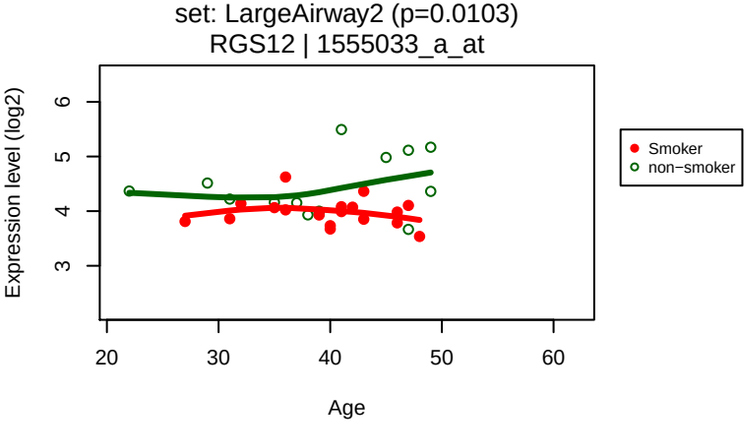

Gene expression ratio (smokers/nonsmokers)
gene: RGS12 | 1555033_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1852
p=0.5138
p=0.1568
p=0.2685
p=0.0103
p=0.0042
p=0.8136
p=0.1363
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

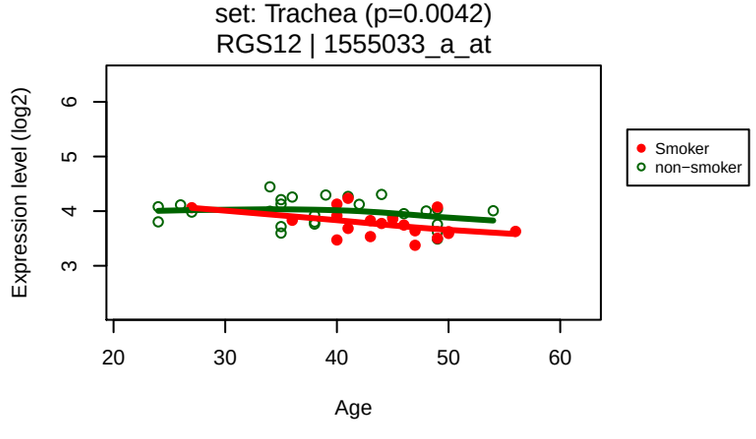

Gene expression ratio in Males
gene: RGS12 | 1555033_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2001
p=0.251
p=0.0769
p=0.2007
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS12 | 1555033_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2163
p=0.0158
p=0.1937
p=0.4834
SmallAirway2
LargeAirway2
Trachea
Macrophage2

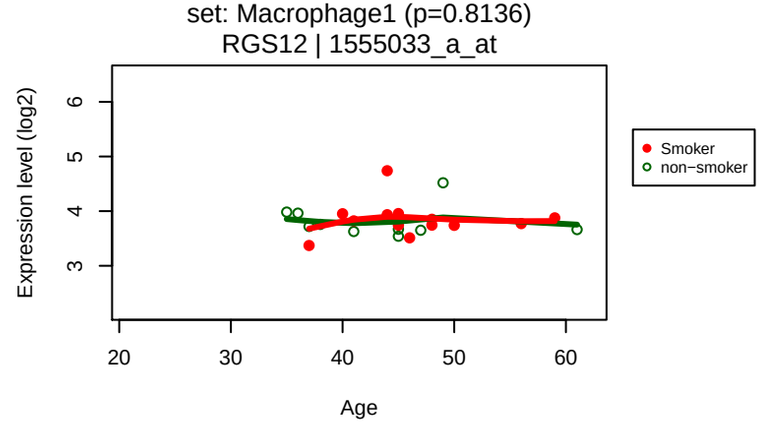

set: SmallAirway1 (p=0.1852)
RGS12 | 1555033_a_at
Expression level (log2)
Age
Smoker
non−smoker

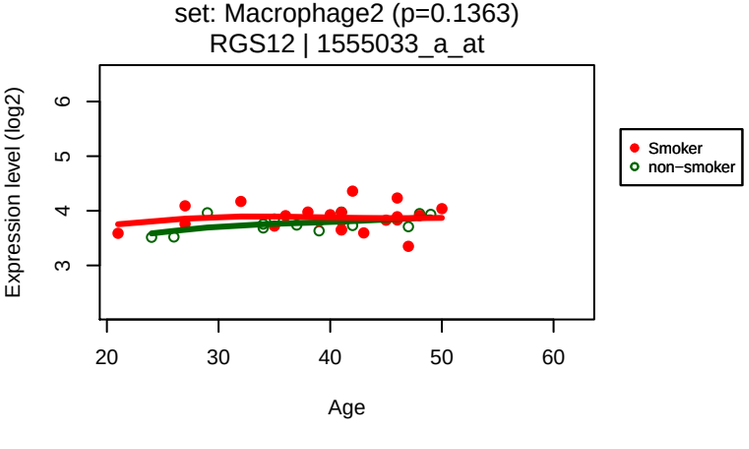

set: SmallAirway2 (p=0.5138)
RGS12 | 1555033_a_at
Expression level (log2)
Age
Smoker
non−smoker

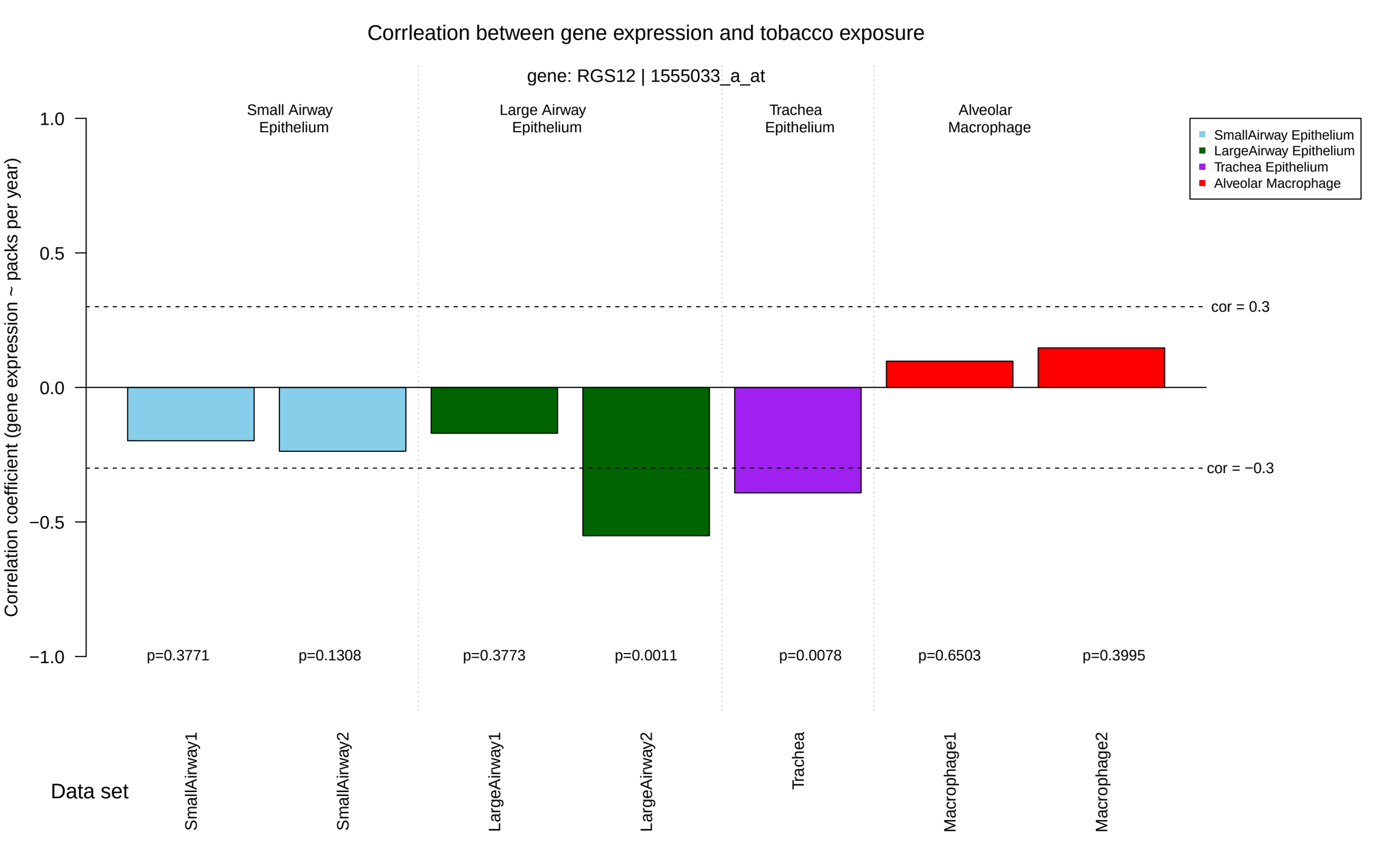

set: LargeAirway1 (p=0.2685)
RGS12 | 1555033_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0103)
RGS12 | 1555033_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0042)
RGS12 | 1555033_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8136)
RGS12 | 1555033_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1363)
RGS12 | 1555033_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS12 | 1555033_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3771
p=0.1308
p=0.3773
p=0.0011
p=0.0078
p=0.6503
p=0.3995
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2